CrowdWisdomTrading: Quantitative Prediction Report

Multi-Horizon Evaluation across 70% Dev & 30% Holdout | Generated 2026-10-20 16:11 UTC

1. Research Methodology & Leakage Prevention

All horizons predict return from confirmed close(t-1) to future close(t+h). Training features use information strictly up to close(t-1). Evaluated using 70% Development (purged walk-forward CV with 5-bar embargo) and 30% Frozen Chronological Holdout.

2. Development Period Walk-Forward Validation Metrics (70% Window)

Horizon	Model	MAE ($)	RMSE ($)	Dir Acc (%)	Base MAE ($)	DM p-val
n+0	Persistence	$184.8	$292.4	49.1%	$184.8	1.000
n+0	Random Forest	$184.9	$292.9	52.3%	$184.8	1.000
n+0	Ridge	$185.2	$292.9	51.4%	$184.8	1.000
n+0	Hist Gradient	$185.8	$293.5	52.0%	$184.8	1.000
n+1	Persistence	$262.6	$406.6	49.1%	$262.6	1.000
n+1	Ridge	$263.6	$407.4	52.7%	$262.6	1.000
n+1	Random Forest	$263.8	$408.3	52.3%	$262.6	1.000
n+1	Hist Gradient	$263.8	$408.9	52.2%	$262.6	1.000
n+2	Persistence	$321.6	$490.8	49.1%	$321.6	1.000
n+2	Random Forest	$322.4	$493.4	52.3%	$321.6	1.000
n+2	Ridge	$323.7	$492.7	50.7%	$321.6	1.000
n+2	Hist Gradient	$323.9	$495.6	52.8%	$321.6	1.000
n+3	Persistence	$369.2	$556.7	50.0%	$369.2	1.000
n+3	Random Forest	$373.6	$562.3	49.9%	$369.2	1.000
n+3	Ridge	$376.3	$562.9	50.9%	$369.2	1.000
n+3	Hist Gradient	$376.8	$566.5	49.9%	$369.2	1.000
n+4	Persistence	$413.7	$614.4	49.9%	$413.7	1.000
n+4	Random Forest	$421.6	$623.5	49.1%	$413.7	1.000

3. Frozen Holdout Validation Metrics (30% Quarantined Window)

Horizon	Model	MAE ($)	RMSE ($)	Dir Acc (%)	Base MAE ($)	DM p-val
n+0	Persistence	$191.2	$291.9	51.0%	$191.2	0.997
n+0	Random Forest	$191.3	$292.0	51.2%	$191.2	0.997
n+0	Hist Gradient	$191.7	$292.8	49.0%	$191.2	0.997
n+0	Ridge	$192.7	$293.6	50.8%	$191.2	0.997
n+1	Persistence	$270.5	$414.0	52.5%	$270.5	0.997
n+1	Random Forest	$271.2	$415.0	50.0%	$270.5	0.997
n+1	Hist Gradient	$273.2	$416.8	51.8%	$270.5	0.997
n+1	Ridge	$274.6	$417.5	50.4%	$270.5	0.997
n+2	Random Forest	$334.5	$503.1	51.2%	$335.3	0.997

n+2	Persistence	$335.3	$502.2	51.3%	$335.3	0.997
n+2	Hist Gradient	$336.5	$505.2	53.2%	$335.3	0.997
n+2	Ridge	$339.9	$508.5	50.6%	$335.3	0.997
n+3	Random Forest	$386.6	$579.1	53.6%	$387.7	0.997
n+3	Persistence	$387.7	$577.3	52.2%	$387.7	0.997
n+3	Hist Gradient	$388.8	$581.5	55.7%	$387.7	0.997
n+3	Ridge	$391.8	$584.3	52.5%	$387.7	0.997
n+4	Random Forest	$436.5	$643.5	53.4%	$436.6	0.997
n+4	Persistence	$436.6	$640.0	52.5%	$436.6	0.997

4. Quantitative Trading Strategy Performance (10 bps Friction)

Model	Horizon	Split	Cum Ret	Exposure	Sharpe	Sortino	Max DD	Trades
No Trade Bas	n+1	Dev_Walk	+0.0%	0.0%	0.00	0.00	0.0%	0
Ridge	n+1	Dev_Walk	-2.5%	4.4%	-1.22	-1.90	-4.8%	128
Random Fores	n+1	Dev_Walk	-1.9%	3.5%	-1.09	-1.90	-9.1%	80
Hist Gradien	n+1	Dev_Walk	-5.1%	3.3%	-2.26	-3.42	-8.3%	84
No Trade Bas	n+1	Frozen_H	+0.0%	0.0%	0.00	0.00	0.0%	0
Ridge	n+1	Frozen_H	-8.4%	6.5%	-6.07	-7.32	-8.5%	64
Random Fores	n+1	Frozen_H	-2.2%	5.4%	-2.30	-3.25	-3.0%	42
Hist Gradien	n+1	Frozen_H	-1.9%	6.3%	-2.00	-2.82	-2.9%	32